

HPY413Assignment: 7

Network traffic monitoring using the Snort intrusion detection system

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Part 5:

The link to the large PCAP file that was used is [Files - Publicly available PCAP files](#). The command used is: `time sudo snort -r /home/student/Desktop/assignment7/snort/lab/maccdc2011_00004_20110311221544.pcap -c /etc/snort/snort.conf -A console`, in order to measure how long Snort takes. The metrics are displayed below:

Time taken:

```
+-----[filtered events]-----
| gen-id=1      sig-id=2923      type=Threshold tracking=dst count=10  seconds=
60 filtered=3626
| gen-id=1      sig-id=2924      type=Threshold tracking=dst count=10  seconds=
60 filtered=10035
Snort exiting

real    2m42.152s
user    0m0.759s
sys     0m6.986s
```

CPU and Memory Usage during execution (htop):

```
CPU[|||                                     3.0%] Tasks: 114, 356 thr, 76 kthr; 1 running
Mem[||||||||| 981M/3.82G] Load average: 1.83 1.85 0.90
Swp[||||| 0K/3.82G] Uptime: 00:05:22
```

A possible suggestion to optimize its performance is to change the above command to the following: `time sudo snort -r /home/student/Desktop/assignment7/snort/lab/slammer.pcap -c /etc/snort/snort.conf -A fast` and the time improved is displayed below:

```
+-----[filtered events]-----
| gen-id=1      sig-id=2924      type=Threshold tracking=dst count=10  seconds=
60 filtered=10035
| gen-id=1      sig-id=2923      type=Threshold tracking=dst count=10  seconds=
60 filtered=3626
Snort exiting

real    0m51.249s
user    0m0.016s
sys     0m0.006s
```